



MALAWI UNIVERSITY OF SCIENCE AND TECHNOLOGY

MALAWI INSTITUTE OF TECHNOLOGY

APPLIED STUDIES DEPARTMENT

2023/24 Mid Semester Examination (ODeL)

PHYS-1101: MECHANICS AND THERMAL PROPERTIES OF MATTER

[50 Marks]

DATE: 2nd April 2024

TIME: 2 hrs (09:00–11:00)

INSTRUCTIONS

1. Write your **Registration Number** on each page of your answer sheets.
 2. **Start each question on a NEW page.**
 3. This paper contains **4 pages** and **3 QUESTIONS**, please check.
 4. Answer **ALL** questions.
 5. Show your work clearly.
 6. Marks allocated for each question are indicated against it.
 7. Round off your final answer to **2 decimal** places.
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LIST OF PHYSICAL CONSTANTS

Acceleration due to gravity	g	9.8 m/s ²
Universal gravitational Constant	G	6.67x10 ⁻¹¹ N•m ² /kg ²

Question one.

A. A projectile is launched from point **A** with the initial conditions shown in the **figure 1**.

- Determine the slant distance s which locates the point **B** of impact. [4 marks]
- Calculate the time of flight t . [8 marks]

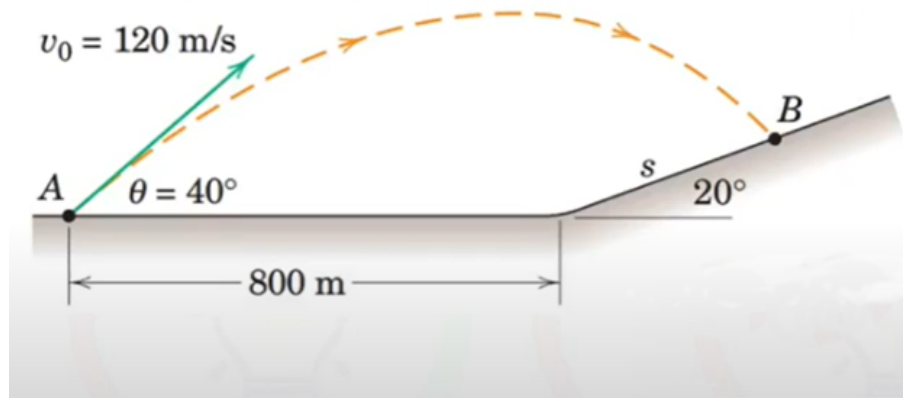


Figure 1

B. A stone that was projected at a cliff of height h with an initial speed of 42.0 m/s directed at an angle 60° the horizontal as **figure 2** shows. The stone strikes at Point **A** 5.50 s after launching. Find the height h of the cliff. [8 marks]

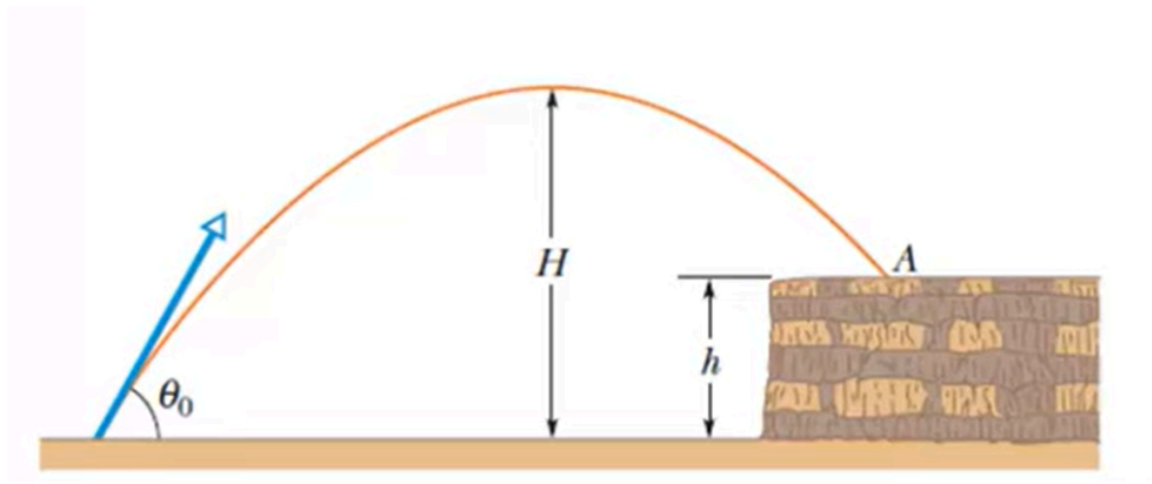


Figure 2

Question 2.

- A. A block with weight W is suspended from cords as shown in **figure 3** below. When the system is in static equilibrium, the tension in the horizontal cord is 15 N. What is the weight W of the block? [6 marks]

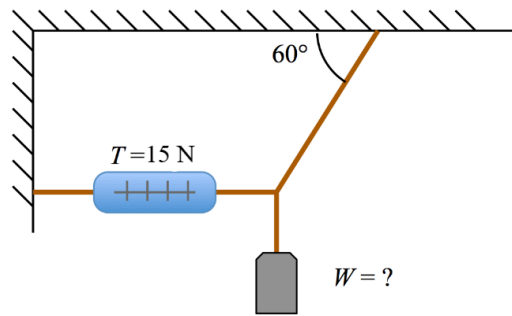


Figure 3

- B. A block of mass m_2 on a rough, horizontal surface is connected to a ball of mass m_1 as shown in **Figure 4**. A force of magnitude F at an angle θ with the horizontal is applied to the block, and the block slides to the right. The coefficient of kinetic friction between the block and surface is μ_k . Determine the magnitude of the acceleration of the two objects. [6 marks]

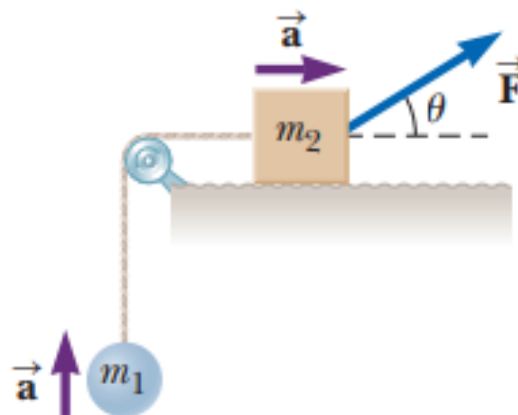


Figure 4

- C. A block of mass m is placed on a plane inclined at an angle $\theta = 30^\circ$ relative to the horizontal as shown in **figure 5**. The coefficient of kinetic friction $\mu_k = 0.15$ acts on the block as it accelerates down the ramp. What is the acceleration of the block?

[6 marks]

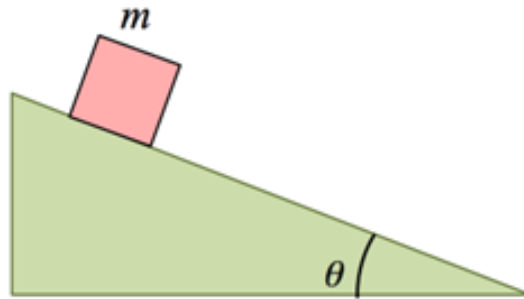


Figure 5

Question 3

- A. An object of mass 2 kg attached to a spring is moved to a distance $x = 10$ m from its equilibrium position. The spring constant $k = 1$ N/m and assume that the surface is frictionless. What is the speed of the mass when it is at the equilibrium position?

[4 marks]

- B. A 10-kg block is placed on a compressed spring and released from rest. The spring was compressed 20 cm and has a spring constant of 11 kN/m. Determine the velocity of the block after it has traveled up the slope for 1.2 m. The coefficient of kinetic friction is 0.4 and $\theta = 45^\circ$.

[8 marks]

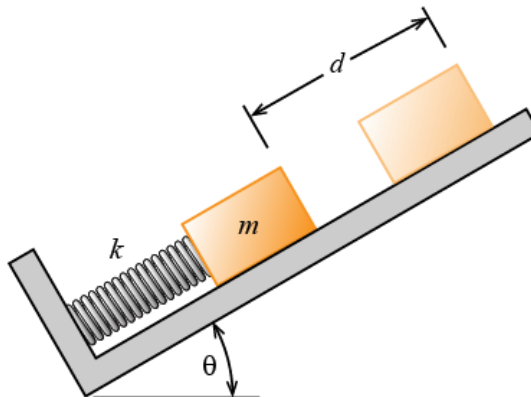


Figure 6

End of Questions